PCB DESIGN

WSN READER

Abstract

This project presents the design and development of a Wireless Sensor Network (WSN) Reader with a focus on the printed circuit board (PCB). The device is designed to collect data wirelessly from data cable and relay it for processing or display. The PCB design emphasizes compactness, low power consumption, and reliable wireless communication.

Key components include a microcontroller, power regulation circuitry, and an RF module (e.g., LoRa). Careful attention was given to component placement, trace width, and overall layout to ensure signal integrity and manufacturability.

TSOTLHE SEIPHEPI

tsotlhenayangseiphepi@gmail.com

“PCB DESIGN : WSN READER AS A CASE STUDY”

INTRODUCTION

This report outlines the design and implementation of a Printed Circuit Board (PCB) for a Wireless Sensor Network (WSN) Parameter Reader. The aim was to develop a standalone system using the STM32F401RETx microcontroller, capable of wirelessly receiving and transmitting sensor data. The schematic and PCB layout were designed and simulated using Proteus .

General Requirement To Run the MCU as a standalone

To operate independently, a microcontroller unit requires several essential components and configurations. These ensure the MCU powers up correctly, runs its program, and communicates effectively with other devices.

- Power source
- Clock source
- Reset circuit
- Boot configuration
- Programming and Debug interface

Peripherals Used

The design of the WSN reader used key peripherals that were integrated to ensure proper functionality and communication with external devices.
These peripherals include:

- UART – this peripheral was used to establish communication between the microcontroller and the Lora wireless module. The UART peripheral was also used as a debugging interface for the microcontroller. This interface used two wires, the Transmit and Receive pins.
- Serial Wire Debug Interface
  This peripheral was implemented using the SWDIO,SWDCLK and SWO wires, which allow for the circuit and microcontroller programming .

- Reset Circuit
  The reset peripheral involves a manual reset button connected to the NRST pin with a 10kΩ pull-up resistor. This circuit ensures that the MCU can be safely restarted during development or in case of faults
- Clock Source
  The MCU's clock is provided by an external 8 MHz crystal oscillator connected to the OSC_IN and OSC_OUT pins, accompanied by two 29pF load capacitors. This clock source provides stable timing essential for precise operation of the MCU and its peripherals.
- General Purpose Input/Output Pins
  GPIO pins were assigned for basic user interaction and status indication

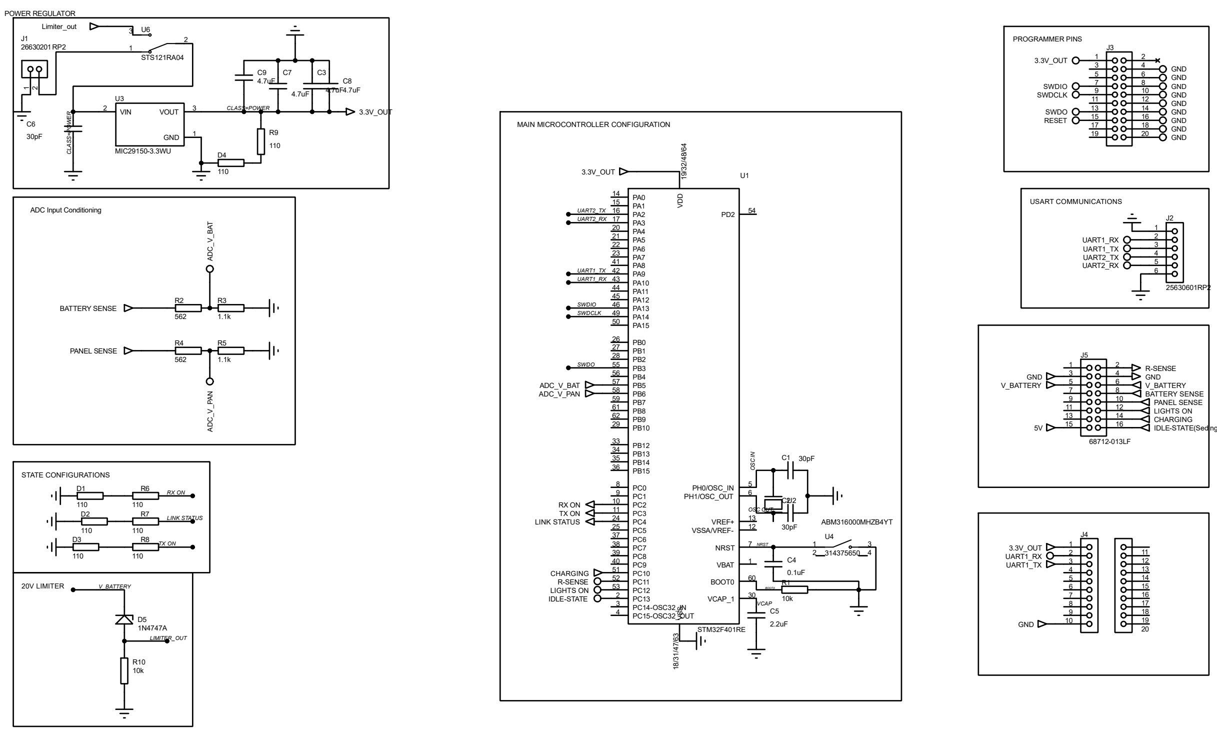

Figure 1: WSN READER CIRCUIT SCHEMATIC

COMPONENT SELECTION

Key factors for component selection when designing this PCB were, availability, precision and stability of components with a particular emphasis on analogue signal integrity.

- **Resistors**: 1% tolerance resistors ensured precise ADC readings; flexibility in values wasn't acceptable.

- **Capacitors**: Low-ESR ceramic capacitors (±10%) were used for decoupling and crystal stability.
- **Crystal Oscillator**: 8 MHz crystal (±20 ppm) ensured stable UART timing and consistent peripheral performance.
- **Power Supply**: A low-noise 3.3V LDO regulator-maintained voltage stability for logic and analog subsystems.
- **Availability**: Components were selected based on local and online availability, with preference for STM32-compatible parts

PCB DESIGN AND LAYOUT

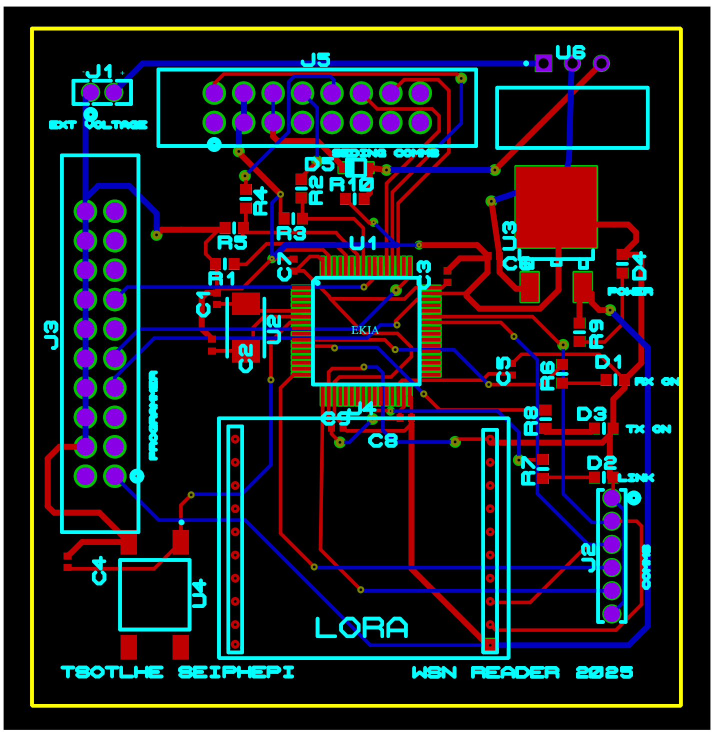

Figure 2: WSN PCB DESIGN LAYOUT

WSN Parameter Reader PCB Layout

- The board size was optimized to 54mm x54 mm.
- All components are placed on the top layer.
- Traces routed to minimize length and avoid interference, especially for power and clock lines.
- SWD , LoRa module connector and power headers accessible on-board edges.

Key Factors Considered for Trace Design

1. Trace Lengths
    - Kept as short as possible to minimize signal delay and losses.
2. Trace Shapes
    - Used gentle 45° bends instead of sharp 90° to reduce signal reflection and impedance discontinuities.
    - Maintained uniform trace width to avoid impedance mismatch and signal degradation.
    - Power traces were sized based on current flow, with wider traces used for higher currents.
3. Via
    - Minimized via count to reduce inductance and resistance.
    - Placed vias only when necessary (e.g., layer transition or routing constraints).
    - Used larger annular rings where high current passed to ensure good conductivity.

Production Files Output

- Gerber files for all PCB layers
- NC drill files for via and hole drilling.
- Bill of Materials (BoM) listing all components with values and footprints.
- Assembly drawings showing component placement.
- Pick and Place files for automated assembly.
- PDF schematic and layout images for documentation.